IGCSE 0607 Practice Questions — E9.2 — Combined Questions & Markscheme

Relative and expected frequencies

Total Questions: 9 — **Total Marks:** 59 Marks

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1 A biased five-sided spinner is numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number.

Number	1	2	3	4	5
Probability	0.15	0.25	x	0.2	$3x$

(a) Work out the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun 250 times.
Calculate the expected number of times the spinner lands on a number greater than 3.

..... **[2]**

2 A biased 4-sided spinner, numbered 1, 2, 3, and 4, is spun 200 times.
The table shows the frequencies of each outcome.

Number	1	2	3	4
Frequency	45	35	70	50

(a) Write down the relative frequency of the spinner landing on 3.

..... **[1]**

(b) The spinner is spun 1000 times.
Estimate the number of times it will land on 1.

..... **[2]**

(c) Is the spinner fair?
Give a reason for your answer.

..... **[1]**

(d) Explain why spinning the spinner 1000 times gives a more reliable estimate of the probability than spinning it 200 times.

..... **[1]**

3 A biased four-sided die is rolled 60 times.
The scores are 1, 2, 3, and 4.
The frequencies of each score are shown in the table.

Score	1	2	3	4
Frequency	x	$2x + 5$	$3x - 2$	$x + 8$

(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find the relative frequency of rolling a 3.

..... **[1]**

(c) The die is rolled 1200 times.
Calculate the expected number of times it rolls a 2.

..... **[2]**

4 A spinner has four colours: Red, Blue, Green, and Yellow.
A student spins it 50 times and records the results.
The table shows the relative frequencies for three of the colours.

Colour	Red	Blue	Green	Yellow
Relative frequency	0.15	0.45	0.10	x

(a) Find the value of x.

$x =$ ..... **[1]**

(b) The spinner is spun 300 times in total.
Estimate the number of times it lands on Blue.

..... **[2]**

(c) The spinner is meant to be fair.
Comment on whether the results show it is fair or biased.

..... **[1]**

5 A bag contains a large number of coloured beads.
A bead is chosen at random, its colour recorded, and then returned to the bag.
This experiment is repeated 300 times.
The table shows the frequencies of the colours drawn.

Colour	Red	Blue	Green	Yellow
Frequency	84	105	x	y

(a) The relative frequency of choosing a Green bead is 0.22.
Work out the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) There are exactly 15 Yellow beads in the bag.
Estimate the total number of beads in the bag.

..... **[2]**

(c) A new experiment is done where a bead is chosen and replaced 800 times.
Calculate the expected number of times a Blue or Green bead is chosen.

..... **[2]**

6 A researcher records the colours of cars passing a checkpoint.
Over one week, 800 cars pass the checkpoint.
The frequencies of the colours are shown in the table.

Colour	Silver	Black	White	Other
Frequency	260	180	w	$w + 40$

(a) Find the relative frequency of a car being Silver.
Give your answer as a fraction in its simplest form.

..... **[2]**

(b) Find the value of w.

$w =$ ..... **[2]**

(c) The researcher uses these results to estimate the probability that the next car passing the checkpoint is White.
Work out this estimate.

..... **[1]**

(d) In one month, 12000 cars are expected to pass the checkpoint.
Calculate the expected number of Black cars.

..... **[2]**

7 A biased 5-sided spinner has sides numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number in terms of x.

Number	1	2	3	4	5
Probability	0.15	$2x$	$x + 0.05$	0.2	x

(a) Find the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun 600 times.
Work out the expected number of times it lands on a prime number.

..... **[2]**

(c) A different spinner has 3 sections: Red, Blue, and Green.
The probability of landing on Red is p.
The probability of landing on Blue is $2p$.
The probability of landing on Green is $1 - 3p$.
This spinner is spun N times.

The expected number of times it lands on Red is 54 and the expected number of times it lands on Green is 36.
Work out the value of N and the value of p.

$N =$ ..... , $p =$ ..... **[4]**

8 A biased four-sided spinner is numbered 1, 2, 3 and 4.
The table shows the probability of scoring each number.
The spinner is spun 400 times.
Let E_1 be the expected number of times the spinner lands on 1.
Let E_3 be the expected number of times the spinner lands on 3.
You are given that $E_1 \times E_3 = 5600$.
Also, the probability of scoring a 3 is greater than the probability of scoring a 1.

Score	1	2	3	4
Probability	x	$2x$	y	y

(a) Show that $300x^2 - 100x + 7 = 0$.

[3]

(b) Solve the equation to find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) Calculate the expected number of times the spinner lands on an even number.

..... **[2]**

9 Two biased coins, Coin A and Coin B, are tossed.
The probability of Coin A landing on Heads is p.
The probability of Coin B landing on Heads is $p + 0.15$.
Coin A is tossed 300 times and Coin B is tossed 200 times.
The total expected number of Heads from these 500 tosses is 255.

(a) Form an equation in p and show that $p = 0.45$.

[3]

(b) Both coins are tossed together 1000 times.
Work out the expected number of times they land on different outcomes.

..... **[3]**

(c) In a new experiment, Coin A is tossed N times.
The relative frequency of landing on Heads is 0.48 and it lands on Heads 72 times.

(i) Work out the value of N.

$N =$ ..... **[2]**

(ii) Calculate the expected number of Tails in these N tosses, using the theoretical probability $p = 0.45$.

..... **[2]**

1 **(a)** $0.15 + 0.25 + x + 0.2 + 3x = 1$ **[M1]**

$x = 0.1$ **[A1]**

(b) $P(> 3) = P(4) + P(5) = 0.2 + 3(0.1) = 0.5$ **[M1]**

Expected number $= 250 \times 0.5 = 125$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $\frac{70}{200}$ or $\frac{7}{20}$ or 0.35 **[B1]**

(b) $1000 \times \frac{45}{200}$ **[M1]**

225 **[A1]**

(c) No, because the relative frequencies are not all close to 0.25 (or 50 out of 200) **[B1]**

(d) A larger number of trials leads to a relative frequency that is closer to the theoretical probability **[B1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $x + 2x + 5 + 3x - 2 + x + 8 = 60$ **[M1]**

$7x + 11 = 60 \implies 7x = 49$

$x = 7$ **[A1]**

(b) Frequency of $3 = 3(7) - 2 = 19$

$\frac{19}{60}$ **[B1]**

(c) Frequency of $2 = 2(7) + 5 = 19$

Relative frequency of $2 = \frac{19}{60}$

$1200 \times \frac{19}{60}$ **[M1]**

380 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $x = 1 - (0.15 + 0.45 + 0.10)$

$x = 0.30$ **[B1]**

(b) 300×0.45 **[M1]**

135 **[A1]**

(c) Biased, because the relative frequencies are not all close to 0.25 (or specifically Blue is 0.45, which is much higher than 0.25) **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $\frac{x}{300} = 0.22 \implies x = 300 \times 0.22$ **[M1]**

$y = 300 - (84 + 105 + 66)$ **[M1]**

$x = 66$, $y = 45$ **[A1]**

(b) Relative frequency of Yellow $= \frac{45}{300}$ or 0.15 **[M1]**

Total beads $= \frac{15}{0.15} = 100$ **[A1]**

(c) Probability of Blue or Green $= \frac{105 + 66}{300} = \frac{171}{300}$ or 0.57 **[M1]**

$800 \times 0.57 = 456$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $\frac{260}{800}$ **[M1]**

$\frac{13}{40}$ **[A1]**

(b) $260 + 180 + w + w + 40 = 800 \implies 480 + 2w = 800$ **[M1]**

$w = 160$ **[A1]**

(c) $\frac{160}{800}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(d) $P(\text{Black}) = \frac{180}{800} = \frac{9}{40}$ **[M1]**

Expected number $= 12000 \times \frac{9}{40} = 2700$ **[A1]**

[Back to Question] *[Total Marks: 7]*

7 **(a)** $0.15 + 2x + x + 0.05 + 0.2 + x = 1$ **[M1]**

$4x + 0.4 = 1 \implies 4x = 0.6 \implies x = 0.15$ **[A1]**

(b) $P(2, 3, 5) = 2(0.15) + (0.15 + 0.05) + 0.15 = 0.3 + 0.2 + 0.15 = 0.65$ **[M1]**

$600 \times 0.65 = 390$ **[A1]**

(c) Sets up both equations: $N \times p = 54$ and $N \times (1 - 3p) = 36$ **[M1]**

Expands the second equation: $N - 3Np = 36$ **[M1]**

Substitutes $Np = 54$ into the expanded equation: $N - 3(54) = 36$ **[M1]**

$N = 198$, $p = \frac{54}{198} = \frac{3}{11}$ (or $0.272...$) **[A1]**

[Back to Question] *[Total Marks: 8]*

8 **(a)** Sum of probabilities: $3x + 2y = 1 \implies y = \frac{1 - 3x}{2}$ **[B1]**

$E_1 = 400x$ and $E_3 = 400y \implies 400x \times 400y = 5600 \implies xy = \frac{7}{200}$ **[M1]**

Substitute for y: $x\left(\frac{1-3x}{2}\right) = \frac{7}{200} \implies 100x(1-3x) = 7 \implies 300x^2 - 100x + 7 = 0$ **[A1]**

(b) Factorising $(30x-7)(10x-1) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = \frac{7}{30}$ and $x = \frac{1}{10}$ (or 0.1) **[A1]**

When $x = \frac{7}{30}$, $y = 0.15$ (reject as $y < x$). When $x = 0.1$, $y = 0.35$ (accept as $y > x$) **[M1]**

$x = 0.1, y = 0.35$ **[A1]**

(c) $P(\text{even}) = 2x + y = 2(0.1) + 0.35 = 0.55$ **[M1]**

Expected number $= 400 \times 0.55 = 220$ **[A1]**

[Back to Question] *[Total Marks: 9]*

9 **(a)** $300p + 200(p + 0.15) = 255$ **[M1]**

$300p + 200p + 30 = 255$ **[M1]**

$500p = 225 \implies p = \frac{225}{500}$

$p = 0.45$ **[A1]**

(b) Probability Coin B is Heads $= 0.45 + 0.15 = 0.60$

Probability of different outcomes = Probability (H, T) + Probability (T, H)

$(0.45 \times 0.40) + (0.55 \times 0.60)$ **[M1]**

$0.18 + 0.33 = 0.51$ **[M1]**

$1000 \times 0.51 = 510$ **[A1]**

(c) **(i)** $\frac{72}{N} = 0.48$ **[M1]**

$N = \frac{72}{0.48}$

$N = 150$ **[A1]**

(ii) Probability of Tails $= 1 - 0.45 = 0.55$

150×0.55 **[M1]**

82.5 **[A1]**

[Back to Question] *[Total Marks: 10]*